

DATA ANALYST TRAINING ACADEMY

Week 9 · Section 5.A: Working with Visualizations

Power BI Desktop — May 2026 Edition

Hands-On Practice Exercise

Building a Complete Power BI Report with 11 Visualization Types

Step-by-step instructions with exact UI locations for Power BI Desktop (May 2026)

Topics Covered

Card · Bar Chart · Column Chart · Line Chart · Area Chart
Pie Chart · Donut Chart · Treemap · Scatter · Bubble · Map

Estimated Time: 90–120 minutes · Skill Level: Beginner–Intermediate

Overview

In this hands-on exercise you will load a provided sample sales dataset into Power BI Desktop and build a two-page report that demonstrates 11 core visualization types. Every step references the exact location of controls in the May 2026 release of Power BI Desktop. By the end you will have a working report you can compare against the completed reference in Section 5.

Learning Objectives

- Choose the correct visualization for a given analytical question
- Locate and configure every field well in the Visualizations pane (Build visual tab)
- Use the Format visual pane (Visual tab and General tab) to add titles, data labels, and colour
- Use on-object formatting handles to resize, move, and align visuals on the canvas
- Interpret expected outcomes and self-assess your work against the reference screenshots

Power BI Desktop Interface — May 2026 Quick Reference

Before you start, make sure you can identify these areas of the interface:

Area	Location	What It Does
Ribbon	Top of screen	Tabs: Home, Insert, Modeling, View, Optimize, Help
Report canvas	Centre	Where you place and arrange visuals
Visualizations pane	Right side	Three inner tabs: Build visual (bar chart icon), Format visual (paint roller), Add further analyses (magnifier)
Data pane	Far right	All tables and columns from your loaded data
View switcher	Left rail icons	Report view (chart), Table view (grid), Model view (diagram)
Page tabs	Bottom bar	Click + to add pages; double-click tab to rename

Tip: If any pane is missing, go to View tab on the ribbon and enable it under Show panes. The Visualizations pane and Data pane are toggled with the View > Panes menu.

Prerequisites

- Power BI Desktop installed — download from: <https://powerbi.microsoft.com/en-us/downloads/>
- After installing, sign in with a Microsoft account (free) or skip sign-in for offline use
- Section 5.A student notes reviewed (especially 5.A.1–5.A.11)

SECTION 1 — SAMPLE DATASET

Sample Dataset: Regional Sales Data

The SalesData.csv file is provided alongside this document. It contains 24 rows of fictional monthly sales data across four US regions, six months, and five products — designed to work well with every visualization type in this exercise.

1.1 Dataset Table

Download the SalesData.csv file and keep it ready for import.

1.2 Data Dictionary

Column	Data Type in Power BI	Description
Month	Text	Calendar month of the row (Jan through Jun)
Region	Text	Sales region: North, South, East, or West
Product	Text	One of five product names: Widget Pro, Gadget X, Cable Set, Monitor 27, Keyboard
Category	Text	Product category: Electronics or Accessories
Units Sold	Whole Number	Number of units sold in that month/region/product combination
Revenue	Fixed decimal number (Currency)	Total revenue for the row before discounts are applied
Discount %	Decimal Number	Discount percentage applied to this transaction (e.g. 5 = 5%)
Latitude	Decimal Number	Geographic latitude of the city — used for precise map pin placement
Longitude	Decimal Number	Geographic longitude of the city — used for precise map pin placement
City	Text	US city name associated with the region row

1.3 Loading the Data into Power BI Desktop (May 2026)

1. **Launch Power BI Desktop.** Close the splash screen or Welcome dialog that appears on startup. You will see a blank Report canvas with no visuals.
2. **Open the data import dialog:** On the Home ribbon, click the Get Data button (top-left group). In the dropdown list that appears, click Text/CSV. A file browser opens.
3. **Select the CSV:** Navigate to the folder where you saved SalesData.csv, select it, and click Open. Power BI displays a preview of the data in a dialog showing the first 200 rows.

4. **Check the delimiter:** In the preview dialog, confirm the Delimiter is set to Comma and the Data type detection is set to Based on first 200 rows. If any column appears as the wrong type, you will fix this in the next step. Click Load.
5. **Verify data types in Table view:** In the left rail, click the Table view icon (grid symbol, second from top). Click the SalesData table in the Data pane on the right. For each column listed below, click the column header and confirm the data type shown in the Column tools ribbon tab matches the expected type. To change a type: click the column header, then on the Column tools tab click the Data type dropdown and select the correct type.
 - a. Revenue → Fixed decimal number (Currency) — look for the \$ symbol next to the column header
 - b. Units Sold → Whole Number — look for the 123 icon
 - c. Discount % → Decimal Number — look for the 1.2 icon
 - d. Latitude → Decimal Number
 - e. Longitude → Decimal Number
 - f. Month, Region, Product, Category, City → Text — look for the ABC icon
6. **Return to Report view:** Click the Report view icon (chart icon, top of left rail) to begin building visuals.
7. **Set up two report pages:** At the bottom of the screen, click the + icon to add a second page. Double-click Page 1 tab and rename it Sales Overview. Double-click Page 2 tab and rename it Distribution & Geography.

If Power BI changed Revenue to a Whole Number automatically, go to Table view, click the Revenue column header, then on the Column tools ribbon set Data type to Fixed decimal number. Click Yes when asked to confirm.

SECTION 2 — BUILDING YOUR VISUALIZATIONS

Report Structure & Layout Plan

Build visuals in the order shown below. Always confirm you are on the correct page (tab at the bottom) before inserting a visual.

Page	Page Name	Visuals to Add (in order)
1	Sales Overview	3× Cards (top row) → Bar Chart (left) → Column Chart (right) → Line Chart (left) → Area Chart (right)
2	Distribution & Geography	Pie Chart → Donut Chart → Treemap (top row) → Scatter Chart → Bubble Chart → Map (bottom row)

Design rule (5.A.1): Choose your visual after deciding on the message. Ask "What question does this chart answer?" — that question becomes the chart title.

Visual 1 Cards — KPI Summary Row

Cards display one large summary number. Placed at the top of a report page they give executives an instant snapshot before they look at the detail charts below. In Power BI Desktop May 2026 the default card visual is the updated "Card (new)" visual with a Callout value and optional Reference labels.

Field Well Configuration

Card	Visual Title	Field	Aggregation
Card 1	Total Revenue	Revenue	Sum (default)
Card 2	Total Units Sold	Units Sold	Sum (default)
Card 3	Avg Discount %	Discount %	Average (must change)

Step-by-Step: Cards

- Go to Page 1 (Sales Overview)** — click the Sales Overview tab at the bottom.
- Click on an empty area of the canvas** so no visual is selected.
- Insert the first Card visual:** In the Visualizations pane on the right, click the Build visual tab (bar chart icon at the top of the pane). Scroll through the visual icons until you find Card — it shows a single large number inside a rectangle. Click it. A blank card placeholder appears on the canvas.
- Add the Revenue field:** In the Data pane (far right), expand the SalesData table. Click and drag the Revenue field into the Fields well (the drop zone labeled "Fields" inside the Visualizations pane Build visual tab). Power BI automatically applies Sum. The card now shows a large dollar amount.
- Change the aggregation if needed:** In the Fields well inside the Visualizations pane, right-click on Revenue. Choose Summarize > Sum (it should already be Sum — verify it is not Count).
- Add the card title:** In the Visualizations pane, click the Format visual tab (paint roller icon). Click General (second inner tab at the top of the Format pane). Expand the Title section. Toggle Title to On. In the Text box, delete any existing text and type: Total Revenue. Set Font size to 12, Font to Segoe UI, and Alignment to Center.
- Style the Callout value:** Click the Visual tab (first inner tab) inside the Format pane. Expand Callout value. Set Font size to 28 and Font to Segoe UI Semibold. Set Display units to Millions if the number is large, or None to show the full value.
- Resize and position the card:** Click the card on the canvas to select it. Drag the resize handles (small squares at the corners and edges) to make the card roughly 3 inches wide and 1.5 inches tall. Drag the card to the top-left area of the canvas.
- Create the second card:** Click an empty area of the canvas to deselect. Click the Card icon in the Visualizations pane again. Drag Units Sold from the Data pane into the Fields well. Title it Total Units Sold (Format visual > General > Title). Position it to the right of the first card at the same height.

10. **Create the third card:** Click empty canvas, add another Card, drag Discount % into Fields. Right-click Discount % in the Fields well and choose Summarize > Average. Title it Avg Discount %. Position it to the right of the second card.
11. **Align all three cards:** Hold Ctrl and click all three cards to select them together. On the Format ribbon tab (appears when visuals are selected), click Align and choose Align top to align them horizontally. Then click Distribute horizontally to space them evenly.

Expected outcome:

- Card 1 shows: \$1,319,850 (or \$1.32M if display units = Millions)
- Card 2 shows: 41,521
- Card 3 shows: 5.5% (or 5.50 if the field is stored as a raw decimal)

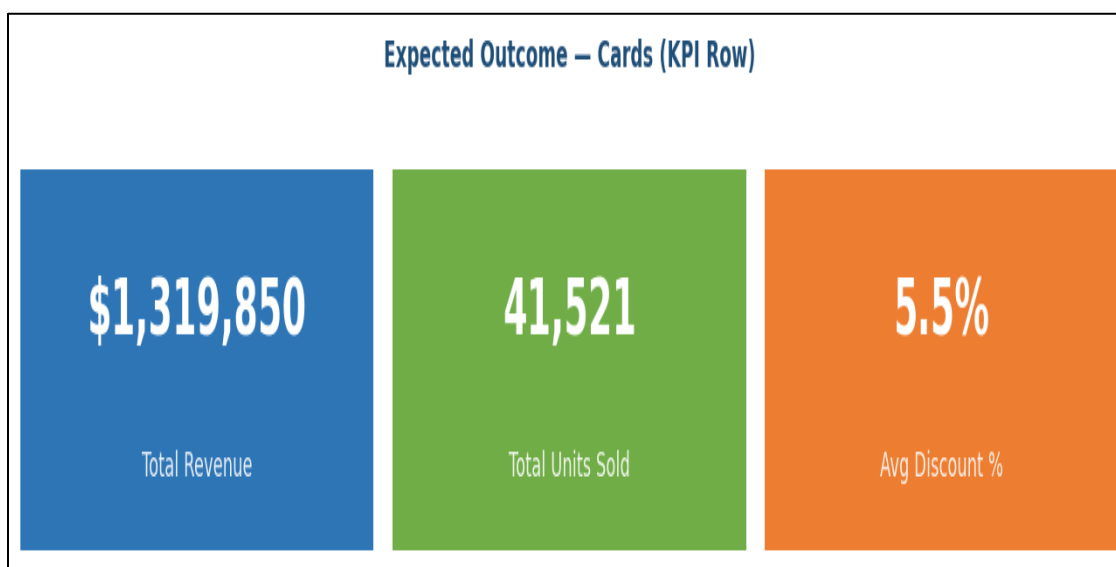


Figure: Expected Outcome — Cards (KPI Row)

✓ May 2026 tip: If you see a "Card (new)" icon and a legacy "Card" icon, use Card (new) for the updated layout. The field well is called "Fields" and the formatting is under Callout value in the Visual tab.

Visual 2 📊 Bar Chart — Total Revenue by Product

A bar chart (horizontal) is best when category labels are long and you want to rank items. Here, five product names appear as horizontal bars sorted by revenue — making it immediately clear which product earns the most.

Field Well Configuration

Field Well	Field	Aggregation	Notes
Y-axis	Product	(categorical — no aggregation)	5 products become the horizontal bars
X-axis	Revenue	Sum	Length of each bar

Legend	(empty)	—	Leave empty; one colour is fine for single-series bars
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Step-by-Step: Bar Chart

1. **Deselect all visuals** by clicking on an empty area of the canvas.
2. **Insert the bar chart:** In the Visualizations pane Build visual tab, click the Clustered bar chart icon (horizontal bars). A blank visual placeholder appears.
3. **Add Product to Y-axis:** In the Data pane, expand SalesData. Drag Product into the Y-axis well in the Visualizations pane. Five horizontal bars appear (one per product, initially all the same length with no values).
4. **Add Revenue to X-axis:** Drag Revenue from the Data pane into the X-axis well. The bars immediately extend to reflect each product's summed revenue.
5. **Sort the bars:** Click the "..." (More options) menu at the top-right corner of the visual on the canvas. Choose Sort axis > Revenue (Sum) to sort by value. Click "..." again and choose Sort descending so the highest bar is at the top.
6. **Enable data labels:** In the Format visual tab (paint roller), click Visual. Expand Data labels. Toggle Data labels to On. Set Font size to 10. Set Display units to Thousands (K) for readability. Set Label position to Outside end so labels appear at the tip of each bar.
7. **Format the bars:** Still in Format visual > Visual, expand Bars. Click the default bar color swatch and choose a dark blue (hex #1F4E79). This distinguishes the bar chart from other visuals that use different colors.
8. **Add the title:** Format visual > General > Title. Toggle On. Type: Total Revenue by Product. Set Font size to 12, Alignment to Left.
9. **Add X-axis label:** Format visual > Visual > X-axis. Toggle On. Set Title to On and type: Revenue (USD).
10. **Resize and position:** Drag the visual to the left-center of the canvas, below the three cards. Make it approximately 4.5 inches wide and 3 inches tall.

Expected outcome:

- Five horizontal bars; Widget Pro and Monitor 27 are the longest
- Bars are sorted longest to shortest (descending)
- Revenue values appear at the right end of each bar

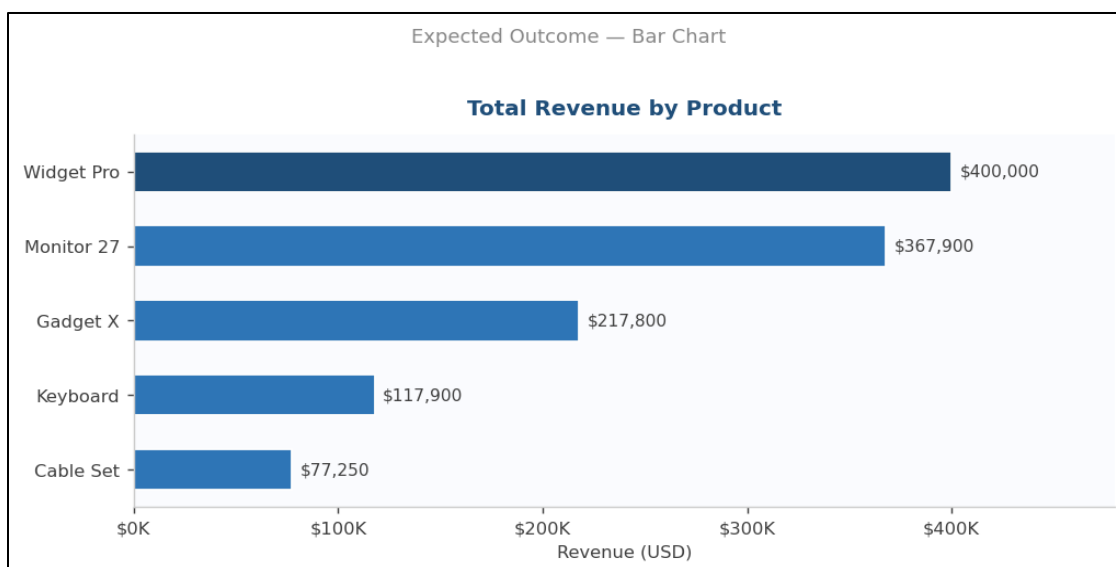


Figure: Expected Outcome — Bar Chart: Total Revenue by Product

⚠ Common mistake: dragging Revenue to Y-axis and Product to X-axis creates a column chart (vertical bars), not a bar chart. If your bars are vertical, swap the fields.

Visual 3 📊 Column Chart — Monthly Revenue by Region

Column charts (vertical) work best when the X-axis is a time series or an ordered sequence with a small number of categories. Here, six months appear across the bottom and four colored column clusters show each region's revenue per month.

Field Well Configuration

Field Well	Field	Aggregation	Notes
X-axis	Month	(categorical)	Jan through Jun across the bottom
Y-axis	Revenue	Sum	Height of each column
Legend	Region	(categorical)	Creates 4 color-coded columns per month group
Small multiples	(empty)	—	Leave empty

Step-by-Step: Column Chart

1. **Deselect all visuals** and click the Clustered column chart icon in the Visualizations pane (vertical bars side by side).
2. **Add Month to X-axis:** Drag Month from the Data pane into the X-axis well. Six month labels appear at the bottom, but the chart is empty because there are no Y-axis values yet.
3. **Add Revenue to Y-axis:** Drag Revenue into the Y-axis well. Tall grey columns appear — one per month, showing the total revenue across all regions.
4. **Add Region to Legend:** Drag Region into the Legend well. Power BI splits each month's bar into four side-by-side colored columns, one per region. A color legend appears automatically.

5. **Fix month sort order:** By default, Power BI may sort months alphabetically (Apr, Feb, Jan...). To fix this: click "..." on the visual > Sort axis > Month. If months are still out of order, you need to create a Month Number column. In the Data pane, right-click SalesData > New column and enter: `MonthNum = SWITCH([Month],"Jan",1,"Feb",2,"Mar",3,"Apr",4,"May",5,"Jun",6)`. Then in the Data pane, click the Month column, go to the Column tools ribbon tab, click Sort by column, and choose MonthNum.
6. **Format the Y-axis:** Format visual > Visual > Y-axis. Toggle On. Set Display units to Thousands. Toggle Title to On and type: Revenue (USD).
7. **Format the legend:** Format visual > Visual > Legend. Toggle On. Set Position to Top center. Set Font size to 10.
8. **Enable data labels (optional):** Format visual > Visual > Data labels > toggle On. Set Font size to 9. Data labels may be crowded on clustered columns; toggle Off if the visual looks cluttered.
9. **Add the title:** Format visual > General > Title > On. Type: Monthly Revenue by Region.
10. **Position the visual:** Place it to the right of the Bar Chart at the same vertical level. Make it approximately 4.5 inches wide.

Expected outcome:

- Six month groups (Jan–Jun) along the X-axis
- Four color-coded columns per group — one per region (North, South, East, West)
- Region legend visible, months in correct calendar order

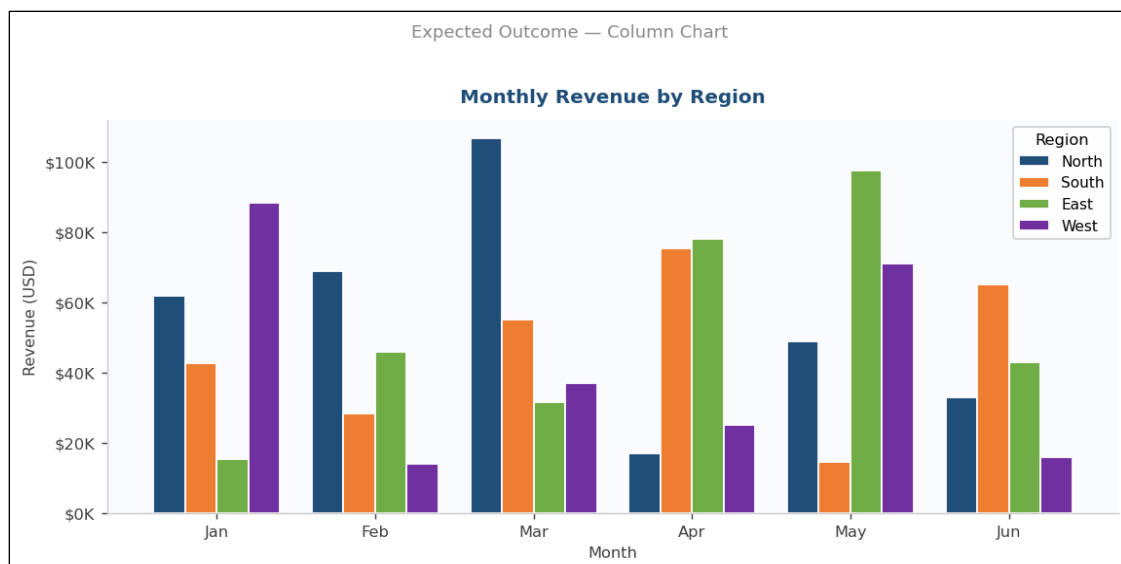


Figure: Expected Outcome — Column Chart: Monthly Revenue by Region

✓ May 2026 tip: Power BI Desktop now offers an "Auto sort" option under Sort axis that intelligently sorts time-based fields. If Month is recognized as a date hierarchy, the sort order will be correct automatically.

Visual 4 Line Chart — Units Sold Over Time by Category

Line charts connect data points with a line to show change over time. They are ideal when you want to show a trend across many time periods and compare two or more series. Here, two lines trace how unit volumes evolved for Electronics and Accessories across the six months.

Field Well Configuration

Field Well	Field	Aggregation	Notes
X-axis	Month	(categorical)	Time axis — Jan to Jun left to right
Y-axis	Units Sold	Sum	Vertical position of each point
Legend	Category	(categorical)	Draws one line per category — Electronics and Accessories
Secondary y-axis	(empty)	—	Leave empty
Tooltips	(optional)	—	Add Revenue here for a tooltip showing revenue on hover

Step-by-Step: Line Chart

1. **Deselect all visuals** and click the Line chart icon in the Visualizations pane (a zigzag line).
2. **Add Month to X-axis:** Drag Month from the Data pane into the X-axis well.
3. **Add Units Sold to Y-axis:** Drag Units Sold into the Y-axis well. A single line appears connecting the monthly totals.
4. **Add Category to Legend:** Drag Category into the Legend well. The single line splits into two lines — one for Electronics and one for Accessories — each coloured differently.
5. **Enable line markers:** Format visual > Visual > Markers. Toggle Markers to On. This places a dot at each data point, making individual month values easy to read.
6. **Adjust line thickness:** Format visual > Visual > Lines. Set Stroke width to 2.5 px for both lines. Under Series, you can select each line individually and change its color.
7. **Format the Y-axis:** Format visual > Visual > Y-axis > Title: On. Type: Units Sold. Set Display units to None (units are already whole numbers).
8. **Enable data labels:** Format visual > Visual > Data labels > On. Set Font size to 9. Set Position to Above to place values above each marker dot.
9. **Add the title:** Format visual > General > Title > On. Type: Units Sold Over Time by Category.
10. **Position the visual:** Place it below the Bar Chart (lower-left area of the canvas). Make it approximately 4.5 inches wide and 3 inches tall.

Expected outcome:

- Two lines trace Jan–Jun: Accessories (higher unit volume, orange line) and Electronics (lower volume, blue line)
- Dot markers visible at each month on both lines
- Both lines show modest variation with no extreme spikes

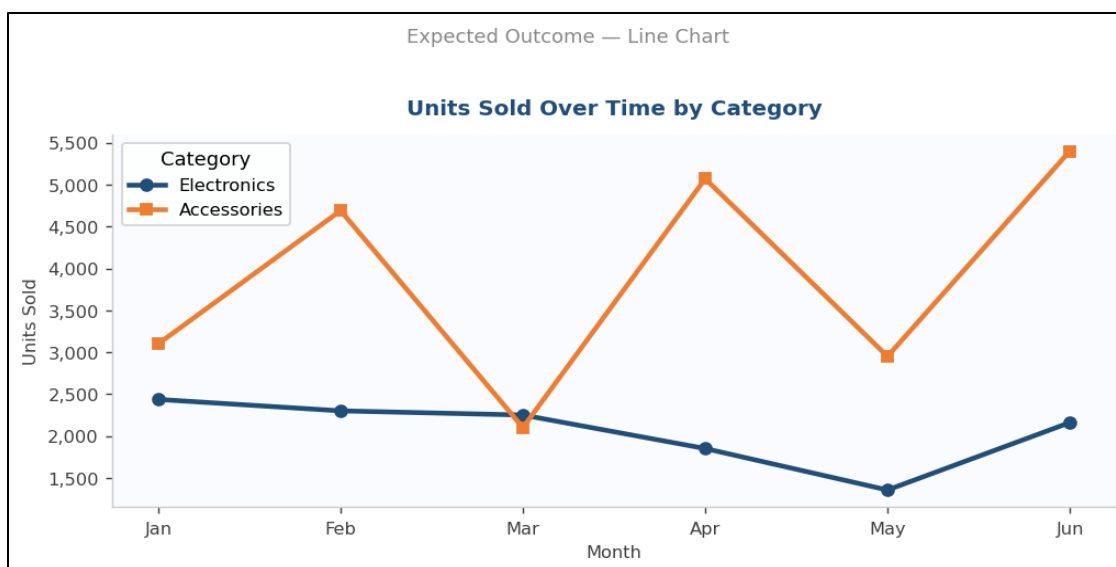


Figure: Expected Outcome — Line Chart: Units Sold Over Time by Category

⚠ If you see only one line, check that Category is in the Legend well (not the X-axis or Y-axis). Without a Legend field, both categories are combined into a single summed line.

Visual 5 Area Chart — Revenue by Category Over Time

An area chart is a line chart where the area below the line is filled. The fill emphasises the cumulative magnitude of each series. Use it here to show how Electronics and Accessories revenue accumulate across the six months.

Field Well Configuration

Field Well	Field	Aggregation	Notes
X-axis	Month	(categorical)	Same time axis as the line chart
Y-axis	Revenue	Sum	Height of the filled area
Legend	Category	(categorical)	Two filled areas — one per category

Step-by-Step: Area Chart

1. **Deselect all visuals** and click the Area chart icon in the Visualizations pane (it looks like a filled line chart — a zigzag with a filled region below it).
2. **Add Month to X-axis:** Drag Month into the X-axis well.
3. **Add Revenue to Y-axis:** Drag Revenue into the Y-axis well. A single filled shape appears.
4. **Add Category to Legend:** Drag Category into the Legend well. Two overlapping filled areas appear — one for Electronics and one for Accessories.
5. **Adjust transparency:** Format visual > Visual > Areas. Set Transparency to 40% so the areas are semi-transparent and you can see both areas where they overlap. Adjust the colour of each series if desired (click a series in the Format pane to change it individually).

6. **Format the Y-axis:** Format visual > Visual > Y-axis > Title: On. Type: Revenue (USD). Set Display units to Thousands.
7. **Enable data labels (optional):** Format visual > Visual > Data labels > On. Set Font size to 9.
8. **Add the title:** Format visual > General > Title > On. Type: Revenue by Category — Area View.
9. **Position the visual:** Place it to the right of the Line Chart (lower-right area of the canvas).

Expected outcome:

- Two semi-transparent filled areas; Electronics sits above Accessories throughout
- The overlapping area shows both categories clearly due to transparency

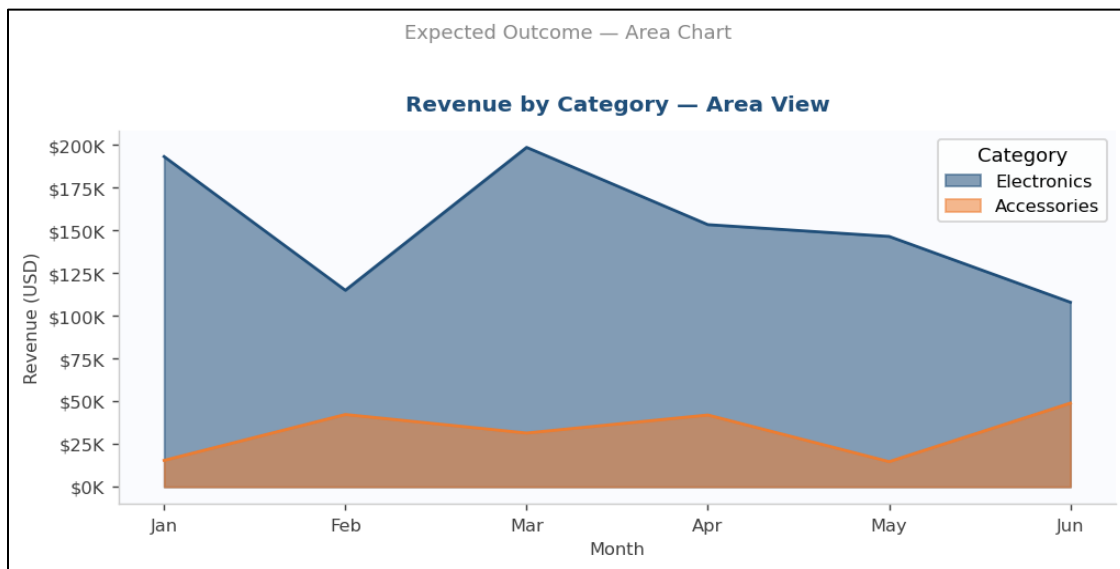


Figure: Expected Outcome — Area Chart: Revenue by Category

✓ May 2026 tip: The Area chart visual in the current Power BI release supports a Stepped area option under Format visual > Visual > Lines > Step. Leave it Off for smooth curves.

Page 2 — Distribution & Geography

Click the Distribution & Geography tab at the bottom before inserting any of the following visuals.

Visual 6 🍷 Pie Chart — Revenue Share by Category

Pie charts show proportional parts of a whole. They work best with 2–5 slices. With only two categories (Electronics and Accessories), a pie chart is an appropriate and readable choice here.

Field Well Configuration

Field Well	Field	Aggregation	Notes
Legend	Category	(categorical)	Creates one slice per category — 2 slices total
Values	Revenue	Sum	Determines the size of each slice
Details	(empty)	—	Leave empty; Details would create one slice per row
Tooltips	(optional)	—	Add Units Sold for a tooltip showing unit volume on hover

Step-by-Step: Pie Chart

1. **Go to Page 2 (Distribution & Geography).**
2. **Deselect all visuals** and click the Pie chart icon in the Visualizations pane (a circle divided into segments).
3. **Add Category to Legend:** Drag Category from the Data pane into the Legend well. Two blank slices of equal size appear.
4. **Add Revenue to Values:** Drag Revenue into the Values well. The slices immediately resize to reflect each category's share of total revenue.
5. **Enable detail labels:** Format visual > Visual > Detail labels. Toggle Detail labels to On. In the Label contents dropdown, select Category and Percent of total. Set Font size to 11. This displays both the category name and the percentage inside or around each slice.
6. **Adjust slice colors:** Format visual > Visual > Slices. Click the first color swatch (Electronics) and set it to dark blue (#1F4E79). Click the second swatch (Accessories) and set it to orange (#ED7D31).
7. **Enable the legend:** Format visual > Visual > Legend > On. Set Position to Bottom center. Font size: 10.
8. **Add the title:** Format visual > General > Title > On. Type: Revenue Share by Category.
9. **Resize:** Make the chart roughly 3 inches wide and 3 inches tall. Place it in the top-left area of Page 2.

Expected outcome:

- Electronics slice ≈ 82% of the pie; Accessories ≈ 18%
- Both labels visible with category name and percentage

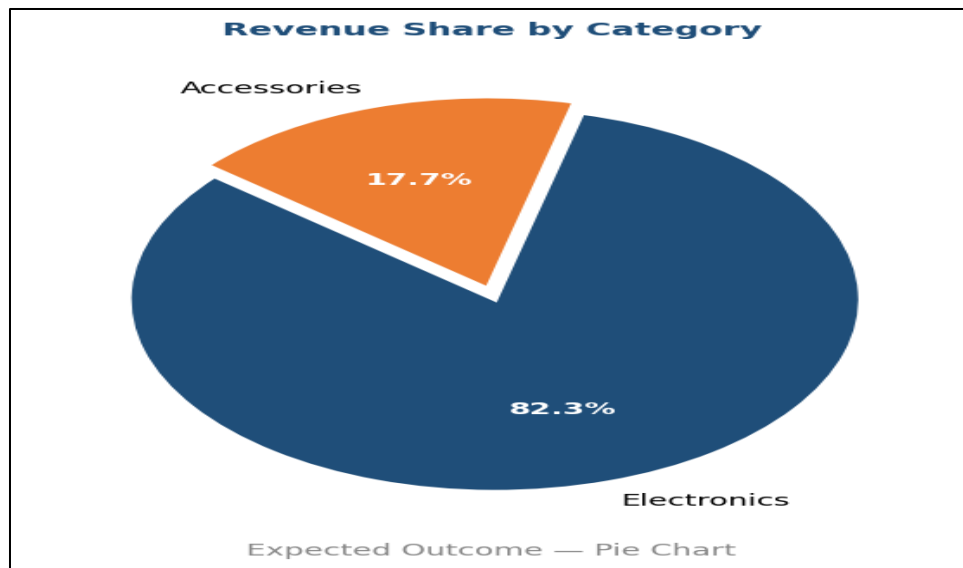


Figure: Expected Outcome — Pie Chart: Revenue Share by Category

⚠ Do NOT put Revenue in the Details well — that would create 24 slices (one per row), making the chart unreadable. Details well should stay empty for a category-level pie chart.

Visual 7 🍩 Donut Chart — Revenue Share by Region

A donut chart is structurally identical to a pie chart but with a hollow centre. The blank centre area can optionally display a summary statistic. Use it here to show four regions' share of total revenue.

Field Well Configuration

Field Well	Field	Aggregation	Notes
Legend	Region	(categorical)	4 slices — one per region
Values	Revenue	Sum	Determines slice size
Details	(empty)	—	Leave empty

Step-by-Step: Donut Chart

1. **Deselect all visuals** and click the Donut chart icon (a circle with a hole in the center).
2. **Add Region to Legend:** Drag Region into the Legend well. Four equal blank slices appear.
3. **Add Revenue to Values:** Drag Revenue into the Values well. Slices resize to reflect each region's revenue share.
4. **Enable detail labels:** Format visual > Visual > Detail labels > On. Label contents: Category and Percent of total. Font size: 10.
5. **Set the inner radius:** Format visual > Visual > Donut. Set Inner radius to 55%. A wider hole gives more breathing room for a center label.
6. **Set slice colors:** Format visual > Visual > Slices. Assign distinct colors to each region (e.g. North = dark blue, South = orange, East = green, West = purple).
7. **Enable the legend:** Format visual > Visual > Legend > On. Position: Bottom center.

8. **Add the title:** Format visual > General > Title > On. Type: Revenue Share by Region.
9. **Position:** Place the donut chart in the top-center area of Page 2, to the right of the Pie Chart.

Expected outcome:

- Four donut slices; North slightly largest, each region contributing roughly 19–30%
- Hollow center clearly visible; legend shows all four region names with color swatches

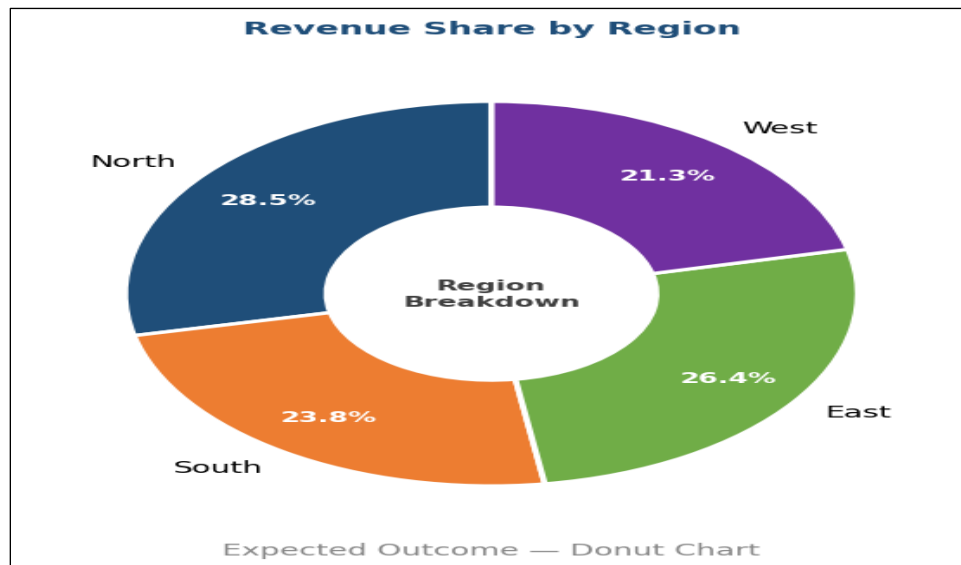


Figure: Expected Outcome — Donut Chart: Revenue Share by Region

✓ To display total revenue inside the donut hole: insert a Text Box (Insert tab > Text box), type "\$1.32M", format it to match the visual, and position it manually in the centre of the ring.

Visual 8 🌳 Treemap — Revenue by Product and Category

A treemap encodes numeric values as rectangle area. Larger rectangles = larger values. Rectangles can be grouped by a category, giving each group its own color. Here, products are the rectangles and categories (Electronics, Accessories) provide the color grouping.

Field Well Configuration

Field Well	Field	Aggregation	Notes
Category	Category	(categorical)	Color group — tiles of the same category share a color family
Details	Product	(categorical)	One rectangle per product within its category group
Values	Revenue	Sum	Rectangle area — larger = more revenue
Tooltips	(optional)	—	Add Units Sold to see volume on hover

Step-by-Step: Treemap

1. **Deselect all visuals** and click the Treemap icon (nested rectangles — it resembles a mosaic of coloured squares).
2. **Add Category to Category well:** Drag Category from the Data pane into the Category well. Two large blocks appear — one for Electronics and one for Accessories.
3. **Add Product to Details well:** Drag Product into the Details well. Each category block is now subdivided into product tiles. You should see 3 tiles in the Electronics group and 2 tiles in the Accessories group.
4. **Add Revenue to Values well:** Drag Revenue into the Values well. Tile sizes immediately reflect each product's revenue. The largest tile (Widget Pro or Monitor 27) appears in the top-left; the smallest (Cable Set) in the bottom-right.
5. **Enable data labels:** Format visual > Visual > Data labels > On. Set Font size to 10. Set Label contents to Name and Value. This displays both the product name and its revenue inside each tile.
6. **Adjust category colors:** Format visual > Visual > Colors. Expand the Electronics group swatch and set to a dark blue family. Expand the Accessories group swatch and set to an orange/warm family.
7. **Enable the legend (category labels):** Format visual > Visual > Category labels > On. This labels each category group with its name (Electronics / Accessories) in a header strip above the tiles.
8. **Add the title:** Format visual > General > Title > On. Type: Revenue by Product and Category.
9. **Position:** Place the treemap in the top-right area of Page 2 to complete the top row.

Expected outcome:

- Widget Pro tile is the largest (highest total revenue \$400K); Cable Set is the smallest (\$77K)
- Electronics tiles appear in one color family; Accessories in another
- Product name and revenue value shown inside each tile

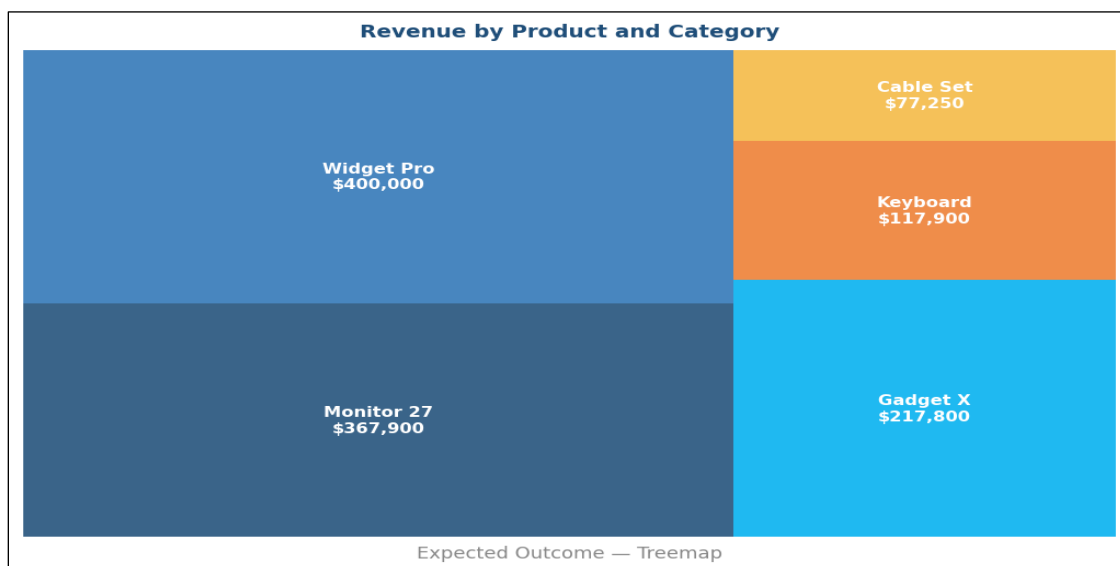


Figure: Expected Outcome — Treemap: Revenue by Product and Category

⚠ If all tiles are the same size, check that Revenue is in the Values well (not the Details well). If Product names are missing, confirm Product is in the Details well (not Values).

Visual 9 ● Scatter Chart — Revenue vs Discount % by Product

Scatter charts plot individual data points on two numeric axes, revealing clusters, correlations, and outliers. Both axes **MUST** be numeric fields — never categorical. Here, each dot represents one product, positioned by its average revenue (X) and average discount percentage (Y).

Field Well Configuration

Field Well	Field	Aggregation	Notes
X-axis	Revenue	Average (change from Sum!)	Horizontal position — avg revenue per product
Y-axis	Discount %	Average	Vertical position — avg discount per product
Values / Details	Product	(categorical — Do Not Aggregate)	Creates one dot per product (5 dots total)
Legend	Category	(categorical)	Colours dots by Electronics vs Accessories
Size	(empty)	—	Leave empty — adding a field here creates a Bubble chart (Visual 10)

Step-by-Step: Scatter Chart

1. **Deselect all visuals** and click the Scatter chart icon (dots scattered across a blank plane — may look like a bubble or scatter icon).
2. **Add Revenue to X-axis:** Drag Revenue into the X-axis well. By default it shows Sum. Right-click Revenue in the X-axis well and change Summarize to Average. This gives each product its average revenue per row rather than its grand total (both work, but Average scales better when products appear in multiple rows).
3. **Add Discount % to Y-axis:** Drag Discount % into the Y-axis well. Change Summarize to Average in the same way.
4. **Add Product to Values/Details:** Drag Product into the Details field well (it may be labelled "Details" or "Values" depending on your build). This creates one dot per product — you should see 5 dots appear on the chart.
5. **Add Category to Legend:** Drag Category into the Legend well. The 5 dots split into two colors — one color for Electronics products (Widget Pro, Gadget X, Monitor 27) and another for Accessories (Cable Set, Keyboard).
6. **Enable data labels:** Format visual > Visual > Data labels > On. Label contents: Product name. Font size: 10. This labels each dot with the product name so viewers can identify points without hovering.
7. **Format the X-axis:** Format visual > Visual > X-axis > Title: On. Type: Avg Revenue (USD). Display units: Thousands. Show title: On.

8. **Format the Y-axis:** Format visual > Visual > Y-axis > Title: On. Type: Avg Discount %. Display units: None.
9. **Enable gridlines:** Format visual > Visual > Gridlines > Horizontal gridlines: On and Vertical gridlines: On. Set line style to Dashed and color to light grey. Gridlines help readers trace values from each dot to the axes.
10. **Add the title:** Format visual > General > Title > On. Type: Revenue vs Discount % by Product.
11. **Position:** Place the scatter chart in the bottom-left area of Page 2.

Expected outcome:

- 5 dots on the chart — one per product
- Monitor 27 is top-right (highest avg revenue, highest avg discount $\approx 10\%$)
- Cable Set is bottom-left (lowest avg revenue, lowest discount $\approx 1.8\%$)
- Electronics dots are one color; Accessories another

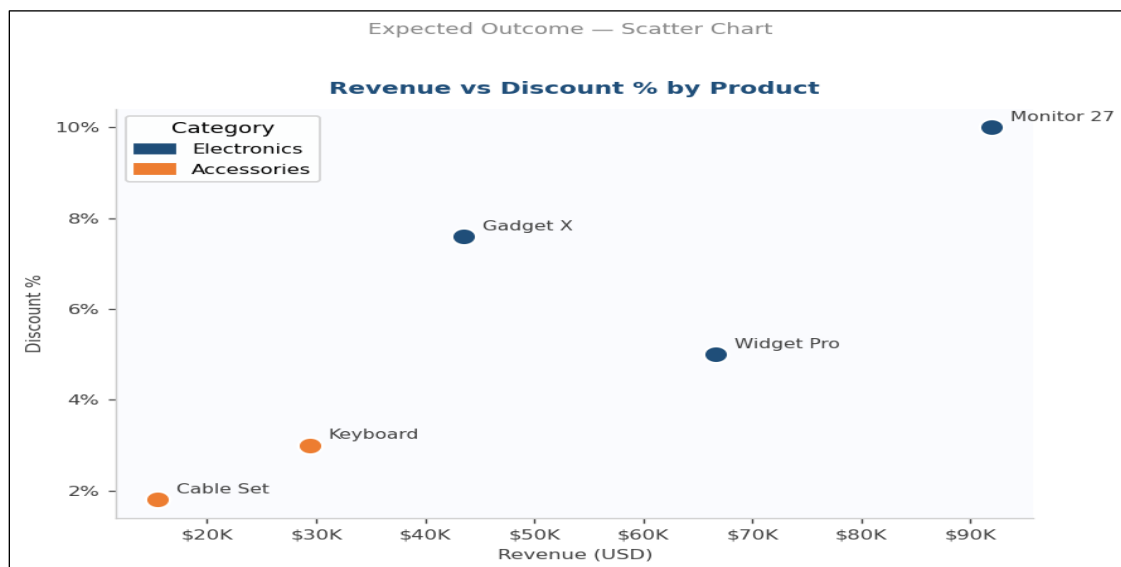


Figure: Expected Outcome — Scatter Chart: Revenue vs Discount % by Product

⚠ If you see many more than 5 dots, Product may not be in the Details well — each row in the dataset would be plotted individually. Make sure Product is in Details, not Revenue or Discount %.

Visual 10 ● Bubble Chart — Revenue, Discount % & Volume

A bubble chart is a scatter chart where a third numeric field controls the bubble size. This encodes three dimensions of data in a single visual. Here the bubble size represents Units Sold, so you can simultaneously see revenue (X), discount (Y), and volume (bubble size) for each product.

Field Well Configuration

Field Well	Field	Aggregation	Notes
X-axis	Revenue	Average	Same as Scatter chart
Y-axis	Discount %	Average	Same as Scatter chart
Size	Units Sold	Average	Controls bubble area — larger bubble = more units sold
Details	Product	(Do Not Aggregate)	One bubble per product
Legend	Category	(categorical)	Colors bubbles by Electronics vs Accessories

Step-by-Step: Bubble Chart

1. **Duplicate the Scatter chart:** Right-click the Scatter chart visual on the canvas and choose Copy. Then right-click an empty area of the canvas and choose Paste. A copy of the scatter chart appears.
2. **Add Units Sold to Size:** With the duplicate visual selected, go to the Visualizations pane Build visual tab. Drag Units Sold from the Data pane into the Size well. The dots immediately transform into bubbles of varying sizes.
3. **Change Size aggregation:** Right-click Units Sold in the Size well and set Summarize to Average.
4. **Enable bubble data labels:** Format visual > Visual > Data labels > On. Labels should show the product name. Adjust Label position to prevent overlap — try Outside or Auto.
5. **Review bubble scaling:** If bubbles are too large or too small, adjust them under Format visual > Visual > Bubbles. Set Min size and Max size to control the range.
6. **Update the title:** Format visual > General > Title > Text. Change to: Revenue, Discount % & Volume by Product.
7. **Add a custom subtitle note:** Format visual > General > Title. Below the main title field, add a Subtitle (if available in your May 2026 build): Bubble size = Units Sold. This helps readers understand the third dimension.
8. **Position:** Place the bubble chart in the bottom-centre of Page 2, to the right of the Scatter chart.

Expected outcome:

- 5 bubbles in the same X/Y positions as the scatter chart
- Cable Set has the largest bubble (highest avg unit volume $\approx 3,110$ units) but sits bottom-left (low revenue)
- Monitor 27 has the smallest bubble (≈ 354 units) but sits top-right (high revenue, high discount)
- Three data dimensions visible simultaneously: position-X (revenue), position-Y (discount), size (volume)

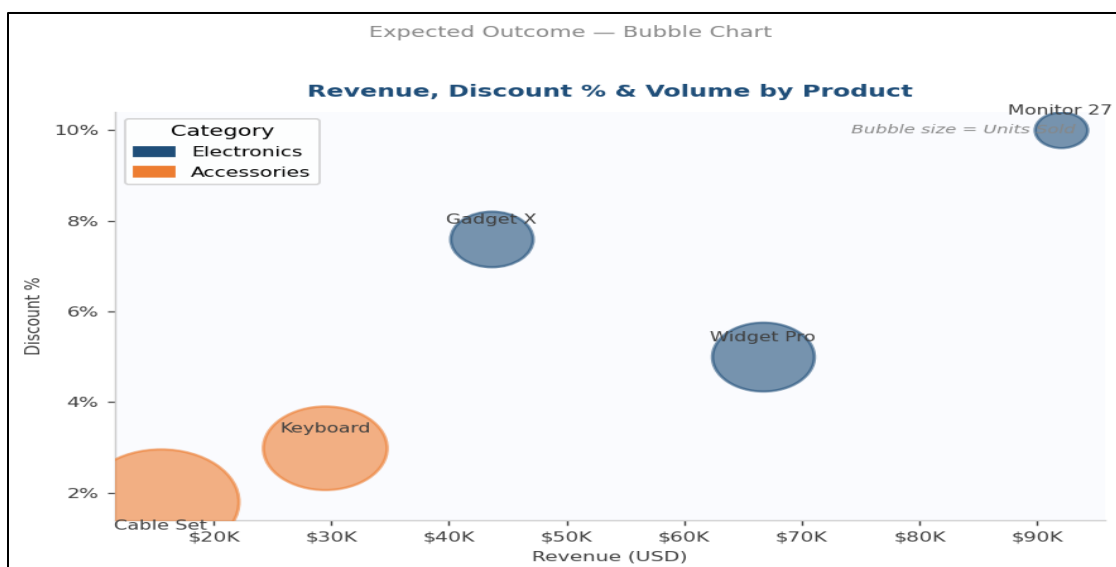


Figure: Expected Outcome — Bubble Chart: Revenue, Discount % & Volume

✓ Bubble charts are powerful for pricing analysis: large bubbles at low prices highlight high-volume, low-margin products (Cable Set), while small bubbles at high prices flag premium low-volume items (Monitor 27).

Visual 11 Map — Revenue by City

Map visuals in Power BI Desktop (May 2026) use Azure Maps as the default engine. Bubble pins are plotted on the map using city name geocoding. Providing explicit Latitude and Longitude columns dramatically improves placement accuracy — especially for cities with names that appear in multiple countries (e.g. "Portland" could be Maine or Oregon).

Field Well Configuration

Field Well	Field	Aggregation	Notes
Location	City	(categorical)	City name — Power BI geocodes this automatically
Latitude	Latitude	Average	Exact latitude overrides geocoding; changes the pin to the correct spot
Longitude	Longitude	Average	Exact longitude paired with Latitude for precision
Bubble size	Revenue	Sum	Controls pin/bubble size — larger = more revenue in that city
Legend	Region	(categorical)	Colour-codes each city pin by its region
Tooltips	(optional)	—	Add Units Sold or Discount % for additional hover detail

Step-by-Step: Map

1. **Enable Map visuals (first time only):** Before inserting a Map for the first time, go to File > Options and settings > Options. In the left panel click Security. Under Map and Filled Map visuals, check the box to enable them. Click OK and restart Power BI Desktop. This is a one-time step.
2. **Deselect all visuals** and click the Map icon in the Visualizations pane (looks like a globe with a pin).
3. **Add City to Location:** Drag City from the Data pane into the Location well. Power BI geocodes the city names and places pins on a map of the United States (since all cities are US cities). You will see roughly 15–20 pins all the same size.
4. **Add Latitude:** Drag Latitude from the Data pane into the Latitude well. Power BI now uses the exact coordinate instead of geocoding the name.
5. **Add Longitude:** Drag Longitude into the Longitude well. Combined with Latitude, pins are now precisely positioned.
6. **Add Revenue to Bubble size:** Drag Revenue into the Bubble size well. Pins grow into proportional bubbles — cities with more revenue show a larger bubble.
7. **Add Region to Legend:** Drag Region into the Legend well. Each city's bubble is now colour-coded by region. A legend appears showing the four region colours.
8. **Enable bubble data labels:** Format visual > Visual > Data labels > On. Label field: City. Font size: 9. This labels each pin with the city name. If too many overlap, toggle Off and rely on hover tooltips instead.
9. **Zoom and frame the map:** On the map visual, use the + and – zoom controls that appear on the map, or use the scroll wheel, to zoom to a level that shows the contiguous United States with all pins visible. Power BI may auto-fit the view when data is first added.
10. **Format the legend:** Format visual > Visual > Legend > On. Position: Top right. Font size: 10.
11. **Add the title:** Format visual > General > Title > On. Type: Revenue by City.
12. **Position:** Place the map in the bottom-right area of Page 2, making it the widest visual on that row (approximately 4–5 inches wide and 3 inches tall).

Expected outcome:

- US map with bubble pins over approximately 20 cities across the country
- New York shows the largest bubble (appears in multiple rows — Jan, Feb, May, Jun — accumulating the highest sum)
- Bubbles are color-coded by region: North (blue), South (orange), East (green), West (purple)
- All pins are correctly placed on the map, not clustered in a wrong location

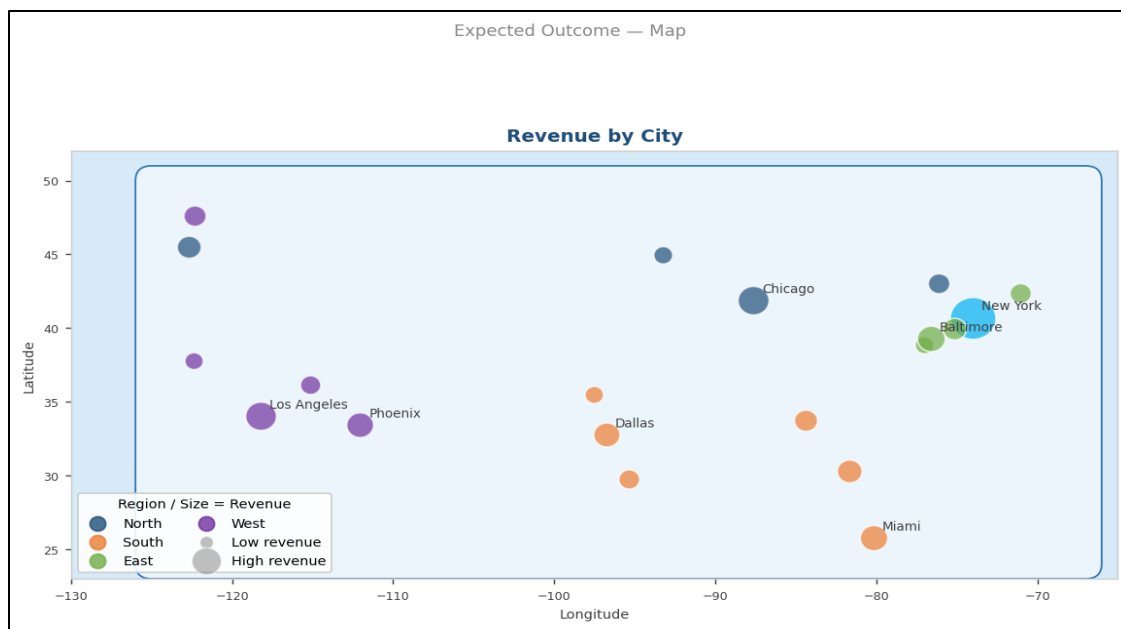


Figure: Expected Outcome — Map: Revenue by City

Note (5.A.10): If a city pin appears in the wrong location (e.g. "Portland" placed in Maine instead of Oregon), the Latitude/Longitude columns will correct it because they take priority over geocoding. Always include lat/long when geographic accuracy matters.

Page 3 — Goals & Performance

Click the Goals & Performance tab at the bottom. Before adding these visuals you need to create two DAX measures to represent target values, since the dataset does not contain a target column.

Creating Target Measures (required for Gauge and KPI)

1. Go to the Home ribbon tab and click Transform data to open Power Query Editor — then close it (you are not transforming data). Instead, stay in Report view and click New measure on the Home ribbon (or right-click SalesData in the Data pane and choose New measure).
2. In the formula bar that appears at the top, type the following measure and press Enter: Revenue Target = 1500000
3. Add a second measure: Units Target = 50000
4. Add a third measure: Monthly Revenue Target = SWITCH(SELECTEDVALUE('SalesData'[Month]), "Jan", 220000, "Feb", 240000, "Mar", 260000, "Apr", 280000, "May", 300000, "Jun", 320000, BLANK())
5. These three measures now appear in the SalesData table in the Data pane and can be dragged into visual field wells like any column.

DAX measure tip: A measure is a formula that is evaluated in the context of the visual using it. Revenue Target = 1500000 always returns 1.5 million regardless of filter context — making it a fixed benchmark you can compare against actual revenue.

Visual 12 Gauge — Revenue Progress Toward Target

A gauge chart shows a single numeric value as a filled arc between a minimum and maximum value. A target line (needle) marks the goal. Gauges are ideal on executive dashboards as a quick "how are we doing?" indicator — used sparingly alongside KPI visuals and cards.

Field Well Configuration

Field Well	Field / Measure	Aggregation	Notes
Value	Revenue	Sum	The current value — fills the arc proportionally
Minimum value	(empty — defaults to 0)	—	Leave empty; Power BI uses 0 as the arc start
Maximum value	Revenue Target (measure)	—	Drag the measure you created; sets the full-arc endpoint at \$1.5M
Target value	(optional — Revenue Target)	—	Draws a target line/needle on the arc at the target amount

Step-by-Step: Gauge

1. **Go to Page 3 (Goals & Performance).**
2. **Deselect all visuals** and click the Gauge chart icon in the Visualizations pane (a semi-circle / speedometer icon).
3. **Add Revenue to Value:** Drag Revenue from the Data pane into the Value field well. The arc fills proportionally, but without a Maximum the gauge defaults to 2× the current value. The arc will appear at 50%.
4. **Set the Maximum value:** Drag the Revenue Target measure into the Maximum value well. The arc now represents progress toward \$1,500,000. At \$1,319,850 actual revenue, the arc fills to approximately 88%.
5. **Set the Target value:** Drag Revenue Target again into the Target value well. A small line (needle) appears on the arc at the \$1,500,000 position, visually marking the goal.
6. **Verify the arc fill:** The arc should be approximately 88% filled ($1,319,850 \div 1,500,000 = 88\%$). If the arc is 50%, your Maximum value well is empty — add Revenue Target there.
7. **Format the callout value:** Format visual > Visual > Callout value. Set Font to Segoe UI Semibold, Size to 24. Set Display units to Millions so the center reads "\$1.32M".
8. **Format the target label:** Format visual > Visual > Target. Toggle Target label to On. Set Font size to 10. The target amount (\$1.5M) appears near the needle.
9. **Format the gauge colors:** Format visual > Visual > Colors. Set Fill color to dark blue (#1F4E79). Set Target color to orange (#ED7D31). Set the remaining arc (empty area) to light grey (#E0E0E0).
10. **Format the data labels (min/max):** Format visual > Visual > Data labels > On. This shows the minimum (\$0) at the left end of the arc and maximum (\$1.5M) at the right end.
11. **Add the title:** Format visual > General > Title > On. Type: Revenue vs \$1.5M Target. Font size: 12.

12. **Position the gauge:** Make it approximately 3 inches wide and 2.5 inches tall. Place it in the top-left area of Page 3.

Expected outcome:

- Arc fills to approximately 88% of the way around
- Orange needle marks the \$1.5M target
- Centre shows \$1.32M in large bold text
- Min (\$0) and Max (\$1.5M) labels visible at the arc endpoints

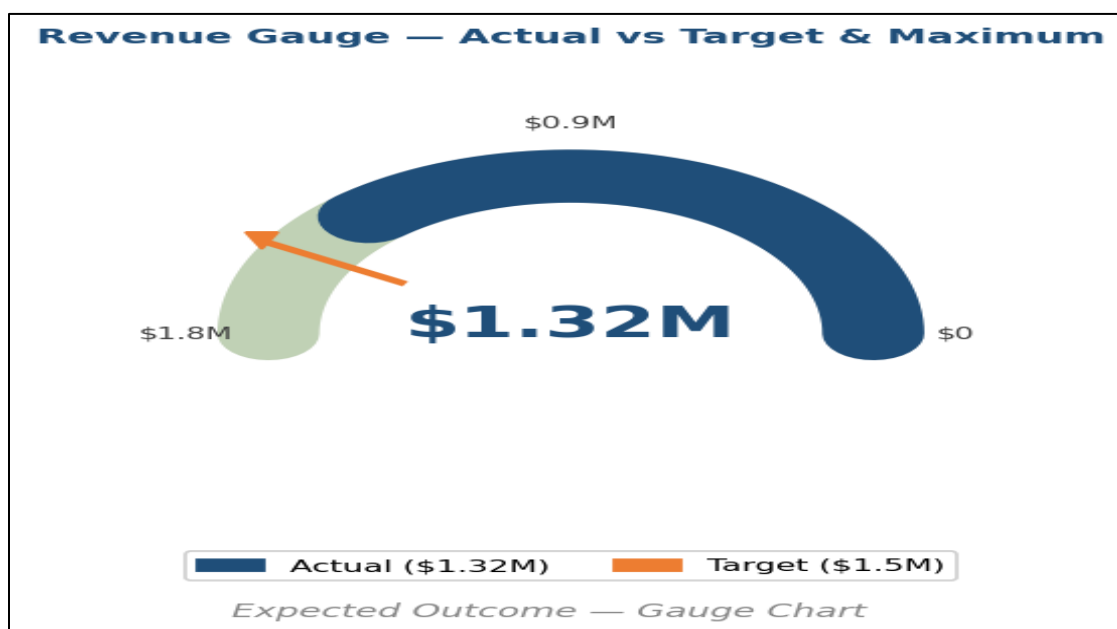


Figure: Expected Outcome — Gauge Chart: Revenue vs Target

⚠ Common mistake: leaving the Maximum value well empty. Without a Maximum, Power BI sets the max to twice the current value — so a \$1.32M revenue with a \$2.64M default max shows the arc at exactly 50%, which is misleading.

Visual 13 KPI Visual — Monthly Revenue vs Monthly Target

The KPI visual in Power BI shows three things at once: (1) the current value, (2) whether it is above or below the target, and (3) a trend over time. It is compact and information-dense — a natural companion to the Gauge on an executive summary page.

Field Well Configuration

Field Well	Field / Measure	Aggregation	Notes
Value	Revenue	Sum	The main metric — shown as the large center number
Trend axis	Month	(categorical)	Draws the spark-line trend at the bottom of the visual

Target goals	Monthly Revenue Target (measure)	—	The benchmark value — determines red/green status color
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Step-by-Step: KPI Visual

1. **Deselect all visuals** and click the KPI icon in the Visualizations pane (it shows a large number with a small trend line beneath it and a coloured status indicator).
2. **Add Revenue to Value:** Drag Revenue into the Value field well. The large number appears in the center.
3. **Add Month to Trend axis:** Drag Month into the Trend axis well. A small spark-line appears at the bottom of the visual, tracing revenue month by month.
4. **Add the target measure:** Drag Monthly Revenue Target into the Target goals well. The KPI now compares each month's actual revenue against the monthly target. The status indicator (typically a coloured arrow) shows whether the current period is above or below target.
5. **Understand the status display:** Power BI uses the last value in the Trend axis to determine status. For June: actual \$157,000 vs target \$320,000 = below target, so the visual shows a red/downward status. This is correct — it accurately shows June underperformed.
6. **Format the callout value:** Format visual > Visual > Callout value. Font: Segoe UI Semibold, Size: 22. Display units: Thousands.
7. **Format the trend line:** Format visual > Visual > Trend line. Set Line color to dark blue (#1F4E79) and Stroke width to 2px.
8. **Format status direction:** Format visual > Visual > Goals. Under Direction, confirm it is set to High is good — this means values above target are shown in green, below in red. If your KPI measures something where lower is better (e.g. cost), set it to Low is good.
9. **Format the goal label:** Format visual > Visual > Goals > Goal label > On. Font size: 10. This shows the target amount below the main value.
10. **Add the title:** Format visual > General > Title > On. Type: Monthly Revenue vs Monthly Target. Font size: 12.
11. **Position:** Make the KPI visual approximately 3.5 inches wide and 2.5 inches tall. Place it to the right of the Gauge on Page 3.

Expected outcome:

- Large number shows total or current-month revenue
- A small spark-line traces the six-month trend beneath the number
- Red status indicator because June actual (\$157K) is below the \$320K June target
- Target amount shown as a label below the main value

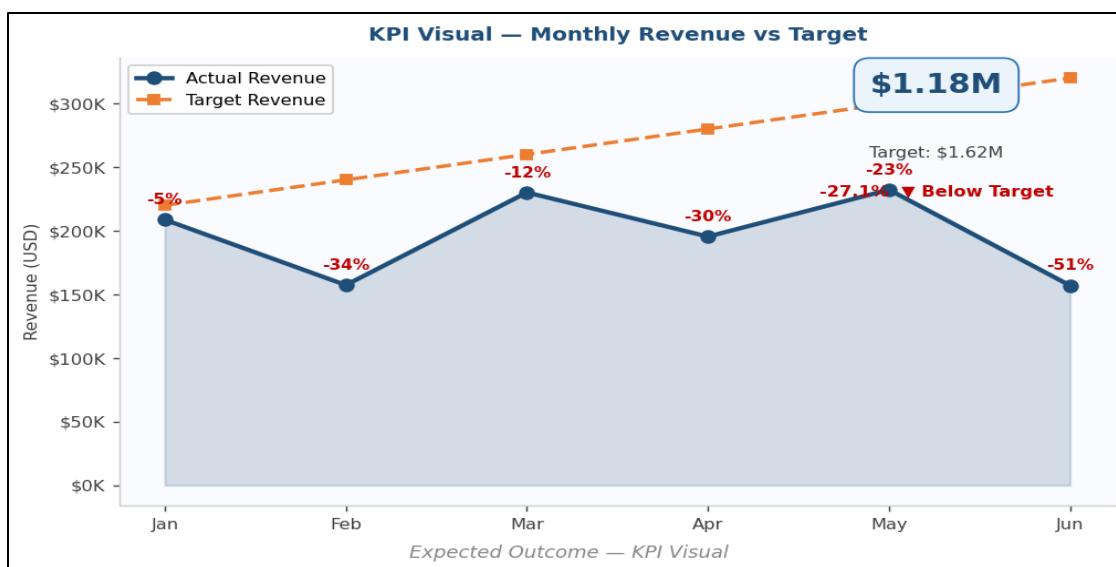


Figure: Expected Outcome — KPI Visual: Monthly Revenue vs Target

✓ May 2026 tip: In recent Power BI releases the KPI visual's Trend axis respects a Sort by column setting. If your months appear out of order in the spark-line, apply the MonthNum sort (created in the Column Chart step) to fix it.

Visual 14 📋 Scorecard (Metrics) — Multi-KPI Performance Table

The Scorecard visual (also called the Metrics visual) displays multiple KPIs in a structured table — each row shows a metric name, its current value, its target, and a status indicator. Unlike the Gauge or KPI visual (which each show one metric), a Scorecard consolidates several metrics for a compact executive overview. In Power BI Desktop May 2026, Scorecards are created and managed through the Metrics hub.

Row	Metric Name	Current Value (Actual)	Target	Expected Status
1	Total Revenue	\$1,319,850	\$1,500,000	Below (red ▼)
2	Total Units Sold	41,521	50,000	Below (red ▼)
3	Avg Discount %	5.5%	≤ 6.0%	On track (green ✓)
4	Electronics Revenue	\$914,700	\$1,000,000	Below (orange ▼)
5	Accessories Revenue	\$195,150	\$250,000	Below (red ▼)

Step-by-Step: Scorecard (Metrics Visual)

- Open the Metrics hub:** On the Insert ribbon tab, click Metrics (also labelled Scorecard in some builds). This opens the Metrics pane on the right side of the screen, or opens a connected Metrics hub dialog.
- Create a new Scorecard:** In the Metrics pane, click + New scorecard (or Connect to scorecard). Name it: Sales Performance Scorecard.
- Add the first metric — Total Revenue:**

- a. Click + Add metric. In the Name field type: Total Revenue.
 - b. Under Connected value, click Select a measure. Choose SalesData > Revenue. Set aggregation to Sum.
 - c. Under Target, click Set target > Manual value. Type: 1500000.
 - d. Under Status rules, confirm the rule is: Value $\geq$ Target = On track (green). Value < Target = Below (red). Click Save.
4. **Add the second metric — Total Units Sold:**
- a. Click + Add metric. Name: Total Units Sold.
 - b. Connected value: SalesData > Units Sold (Sum).
 - c. Target: Manual value = 50000.
 - d. Status rules: same as above (high is good). Click Save.
5. **Add the third metric — Avg Discount %:**
- a. Click + Add metric. Name: Avg Discount %.
 - b. Connected value: SalesData > Discount % (Average).
 - c. Target: Manual value = 6.
 - d. Status rules: Change to Value $\leq$ Target = On track (green). Value > Target = Off track (red). This is a "low is good" metric — a discount above 6% is bad. Click Save.
6. **Add remaining metrics:** Repeat the process for Electronics Revenue (target: 1000000) and Accessories Revenue (target: 250000). For both, filter by Category using the Filter dimension option in the metric configuration to show only the relevant category's revenue.
7. **Insert the Scorecard visual on the canvas:** After configuring all metrics in the Metrics hub, drag the Scorecard visual icon from the Visualizations pane onto the canvas (or click it). The visual automatically connects to the scorecard you just created.
8. **Resize and format:** Make the Scorecard approximately 6 inches wide and 3 inches tall so all five metric rows are visible without scrolling. Format visual > Visual > Grid > Row height: Comfortable.
9. **Format the header row:** Format visual > Visual > Column headers. Set Background color to dark blue (#1F4E79), Font color to white, and Font size to 11.
10. **Format status indicators:** Format visual > Visual > Status. Confirm the icons used for each status: a filled circle, a triangle, or an arrow — choose the style that matches your report theme.
11. **Add the title:** Format visual > General > Title > On. Type: Sales Performance Scorecard.
12. **Position:** Place the Scorecard in the lower half of Page 3, spanning the full width of the canvas.

Expected outcome:

- Five metric rows in a clean table layout
- Total Revenue: \$1,319,850 vs \$1,500,000 — red status (below target)
- Total Units Sold: 41,521 vs 50,000 — red status (below target)
- Avg Discount %: 5.5% vs $\leq 6.0\%$ — green status (within acceptable range)
- Electronics and Accessories Revenue rows both showing below-target red status

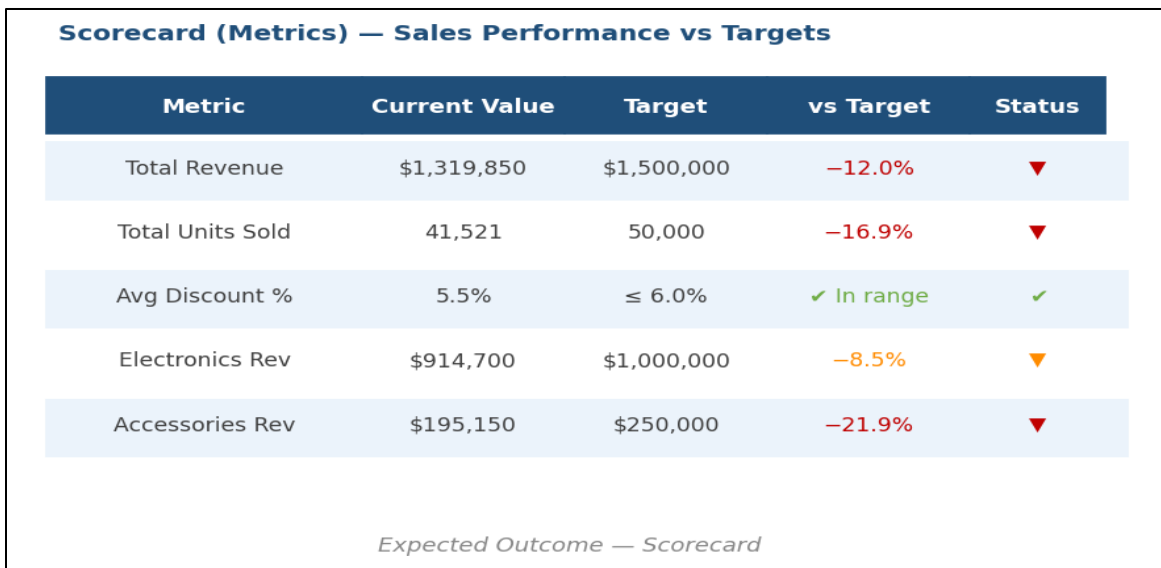


Figure: Expected Outcome — Scorecard: Sales Performance vs Targets

5.A.12 guidance: Use Gauges, KPI visuals, and Scorecards as headline indicators on summary pages — not as primary analytical tools. Pair them with the detail charts (bar charts, line charts, treemaps) on other pages so viewers can explore the "why" behind each status.

SECTION 3 — FORMATTING & LAYOUT BEST PRACTICES

Formatting Your Report (May 2026)

A well-formatted report is as important as the correct visualization choice. Use the guidelines below to make both pages look professional.

3.1 Applying a Report Theme

1. On the View ribbon tab, click the Themes dropdown (top-left group). Browse built-in themes — "Executive", "Classic", and "Colorblind safe" are good starting points.
2. To customize: click Customize current theme. Adjust Primary colour (used for bars, lines, slices) and Secondary color. Click Apply to all visuals.
3. All 11 visuals update simultaneously — much faster than formatting each individually.

3.2 Consistent Titles

- Every visual must have a title: Format visual > General > Title > On
- Use 12pt Segoe UI Bold for all titles
- Titles should be descriptive questions or statements — not just the field name
- Good example: "Which product generates the most revenue?" or "Total Revenue by Product"
- Poor example: "Revenue" or "Chart 1"

3.3 Aligning Visuals

1. Hold Ctrl and click to select multiple visuals.
2. On the Format ribbon tab, use the Align buttons: Align left, Align right, Align top, Align bottom.
3. Use Distribute horizontally or Distribute vertically to space visuals evenly.
4. Use the Selection pane (View > Selection) to rename, reorder, and group visuals.

3.4 Cross-Filtering

This is automatic in Power BI — clicking any data point in any visual automatically filters all other visuals on the same page. For example, clicking the "Electronics" slice in the Pie Chart will highlight only Electronics data in the Bar Chart, Column Chart, and all other visuals on Page 2. No configuration is needed.

3.5 On-Object Formatting (May 2026 Feature)

In May 2026, Power BI Desktop supports on-object formatting. When you select a visual and hover over its title, axis, or legend, small formatting handles appear directly on the visual. Click these handles to open a mini formatting menu without needing to navigate the Format pane. This speeds up common changes like font size and color.

SECTION 4 — EXPECTED OUTCOMES & TROUBLESHOOTING

Expected Outcomes Reference

Visual	Key Numbers to Verify	If It Looks Wrong — Check This
Card — Total Revenue	\$1,319,850	Aggregation = Sum. Revenue field must be Currency type. Not Count.
Card — Total Units Sold	41,521	Units Sold field in well, not Revenue. Aggregation = Sum.
Card — Avg Discount %	5.5% (or 5.50)	Right-click field in well > Summarize > Average (not Sum).
Bar Chart	5 horizontal bars; Widget Pro \$400K longest; Cable Set \$77K shortest	Product on Y-axis, Revenue on X-axis. Clustered bar (horizontal), not column (vertical).

Column Chart	6 month clusters; 4 coloured columns per group	Region in Legend well. Month on X-axis. Months in calendar order (use Sort by column if wrong).
Line Chart	2 lines (Electronics and Accessories) Jan–Jun with dot markers	Category in Legend. Units Sold on Y-axis. Markers enabled in Format pane.
Area Chart	2 filled semi-transparent areas; Electronics higher throughout	Same fields as Line Chart but Area chart icon used. Transparency set to ≈40%.
Pie Chart	2 slices: Electronics ≈82%, Accessories ≈18%	Category in Legend, Revenue in Values. Details well must be EMPTY.
Donut Chart	4 slices; hollow center visible; roughly 19–30% each region	Region in Legend. Inner radius set to ≈55% in Format pane.
Treemap	5 tiles; Widget Pro largest; Cable Set smallest; color-grouped by category	Category in Category well; Product in Details well; Revenue in Values well.
Scatter Chart	5 dots; Monitor 27 top-right; Cable Set bottom-left; both axes numeric	Revenue on X-axis (Average); Discount % on Y-axis (Average); Product in Details.
Bubble Chart	Same as scatter + bubble sizes vary; Cable Set largest bubble	Units Sold in Size well (Average). Confirm 5 bubbles total, not more.
Map	US map; ~20 city pins; sizes proportional to revenue; colour-coded by region	City in Location; Latitude in Latitude well; Longitude in Longitude well; Revenue in Bubble size.

SECTION 5 — COMPLETED REPORT FOR COMPARISON

Completed Report: Page 1 — Sales Overview

The image below shows Page 1 of the fully built and formatted reference report. Compare your layout, visual types, field configuration, and titles against this reference.

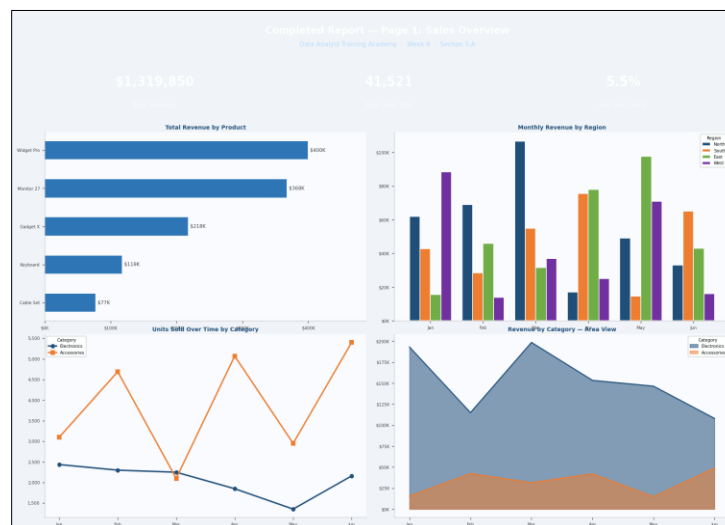


Figure: Completed Report — Page 1: Sales Overview

Completed Report: Page 2 — Distribution & Geography

The image below shows Page 2 of the completed reference report.

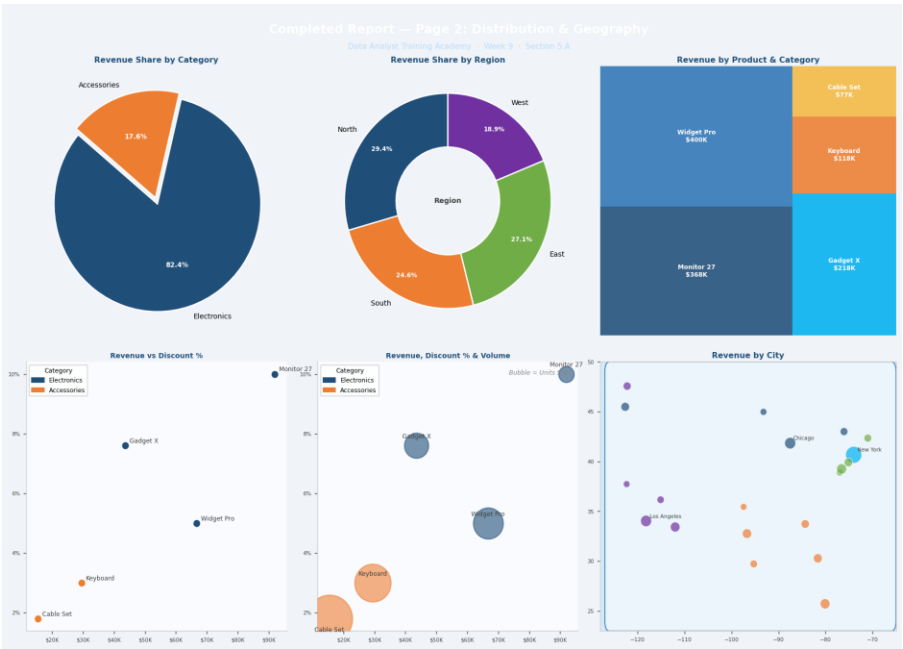


Figure: Completed Report — Page 2: Distribution & Geography

Completed Report: Page 3 — Goals & Performance

The image below shows Page 3 of the completed reference report. Use it to verify the layout and configuration of your Gauge, KPI Visual, and Scorecard.

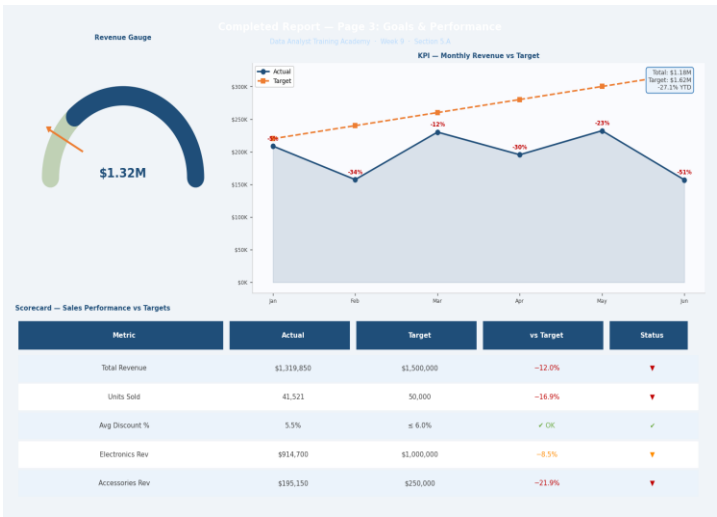


Figure: Completed Report — Page 3: Goals & Performance

SECTION 6 — SELF-ASSESSMENT CHECKLIST

Student Self-Assessment Checklist

☐	Task
☐	Power BI Desktop (May 2026) installed and opened
☐	SalesData.csv loaded via Home > Get Data > Text/CSV
☐	All column data types verified in Table view (Revenue = Currency, Units Sold = Whole Number, Lat/Lon = Decimal)
☐	Page 1 renamed to: Sales Overview
☐	Page 2 renamed to: Distribution & Geography
☐	Card 1 (Total Revenue) shows \$1,319,850 — aggregation = Sum
☐	Card 2 (Total Units Sold) shows 41,521 — aggregation = Sum
☐	Card 3 (Avg Discount %) shows ≈5.5% — aggregation = Average
☐	All three cards aligned horizontally at the top of Page 1
☐	Bar chart — horizontal bars, 5 products, sorted descending by revenue, data labels on
☐	Column chart — 6 month clusters, 4 region colours, months in calendar order
☐	Line chart — 2 lines by category, dot markers enabled, Y-axis = Units Sold
☐	Area chart — 2 filled semi-transparent areas, same data as line chart
☐	Pie chart — 2 slices with % labels, Details well is empty
☐	Donut chart — 4 region slices, hollow centre visible, inner radius ≈55%
☐	Treemap — 5 tiles, Widget Pro largest, Category in Category well, Product in Details well
☐	Scatter chart — 5 dots (one per product), both axes numeric, product name labels on
☐	Bubble chart — same as scatter plus bubble size = Units Sold, Cable Set has largest bubble
☐	Map — Azure Maps, city pins plotted, bubble size = Revenue, pins colour-coded by Region
☐	All 11 visuals have descriptive titles
☐	A report theme has been applied (View > Themes)
☐	Visuals are aligned using Format > Align on both pages
☐	Cross-filtering tested: clicking a data point filters other visuals on the same page
☐	Report compared against Section 5 reference screenshots
☐	Page 3 renamed to: Goals & Performance
☐	DAX measures created: Revenue Target, Units Target, Monthly Revenue Target
☐	Gauge — arc fills to ≈88%; Revenue Target in Maximum value well; orange needle visible

☐	KPI Visual — Monthly Revenue in Value; Month in Trend axis; Monthly Revenue Target in Target goals
☐	Scorecard — 5 metric rows configured in Metrics hub; correct status colours showing

Reflection prompt: Which visualization type was most challenging to configure and why? Which visual do you think most clearly communicates a business insight from this dataset? Share your answer with your instructor or study group.